



## African swine fever: A growing global threat

The global pig industry is worth some US\$150 billion.

### Pig keeping is the most rapidly increasing livestock sector in Africa

In Uganda, pig numbers have increased from 100,000 in the 1970s to 4 million today and continue to rise, and pork consumption is now close to that of beef (FAO data). There were 20 recorded outbreaks of African swine fever in Uganda in 2010 alone. And this is certainly an underestimate due to difficulty in diagnosing the disease and under-reporting of livestock diseases.

Although the absolute total number of pigs kept in Africa remains relatively small (less than 50 million), pig keeping is very profitable for many of Africa's rural poor, providing a flexible means of generating an income in the right environments.

### Impacts of African swine fever in Africa

The Africa-wide economic impacts of African swine fever are hard to quantify due to a dearth of disease recording, especially as this disease rapidly turns lethal in naïve pig herds and active surveillance for the disease is rare. In addition, the prevalence of the disease has thwarted investment in the smallholder pig sector, and those economic impacts are hard to capture.

The disease is still emerging in Africa. In the last 20 years, it has spread to parts of West Africa, Madagascar, Mauritius and most recently (in 2011) to Chad (from Cameroon).

### Impacts of African swine fever in developed countries

It took 35 years (and millions of slaughtered pigs) to eradicate African swine fever from the Iberian Peninsula after the disease spread to Portugal from Angola in 1960 through feeding discarded airline meals (swill) to pigs.

The disease spread rapidly to the Russian Federation following its initial introduction to Georgia from Southeast Africa in 2007 (probably via a ship-assisted pork product). Estimated economic losses in the Russian Federation in 2009 alone were US\$1 billion, with 48,000 pig deaths (OIE data). The spread of the disease to the Ukraine directly threatens pig production in the European Union.

### Impacts of African swine fever in emerging economies

500 million animals are at risk in China, mostly from backyard production by small-scale producers, due to the increasing economic links between Africa and China, which encourages spread of the disease.

### Present solutions

There are no commercially available drugs or vaccines for this disease, so slaughter is the only way to stop outbreaks from spreading.

#### *Biosecurity measures for prevention and mitigation*

ILRI is working with Australia's CSIRO (Commonwealth, Scientific and Industrial Research Organisation) and AusAID as well as, in Spain, CISA (Centro de Investigación en Sanidad Animal / Animal Health Research Center)-Inia (Instituto Nacional de Investigación y Tecnología Agraria y Alimentaria), in partnership with the United Nations Food and Agriculture Organization (FAO), to enhance biosecurity implementation and rapid detection of outbreaks in the field in Africa as interim control measures.

### Potential solutions

#### *Vaccines*

Experimental live attenuated vaccines already exist, providing proof of concept for the feasibility of developing a vaccine against this disease, but these experimental vaccines need to be improved (for better safety and to scale up production) before field deployment is authorized.

Recombinant vaccines may be feasible, but work on these is at an early stage of identifying antigens and best-bet delivery systems.

ILRI has just been awarded major funding from BMZ for vaccine development in collaboration with FLI (Riems) Germany.

#### *Genetically resistant animals*

African wild and domestic pigs exhibit tolerance/ resistance to the African swine fever virus. Research in this area, with the ultimate goal of generating disease-resistant and productive domestic pigs, is just beginning. ILRI is well placed to lead this approach due to its location in an endemic area in Africa and its access to wildlife populations.

### *Drugs*

No drugs are currently in the pipeline and there exists no commercial sector investment in this.

### *Impact of improved control*

More cost-effective prophylactic control through a combination of vaccination, use of genetically disease-resistant pigs and improved management has the potential both to mitigate the impacts of this disease on small-scale farmers in Africa and to reduce the threat to global pork production,

and hence global food security. Another benefit will be minimizing the requirement for the mass slaughter of infected pigs.

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